Ali Alfatemi

Ph.D. Candidate · AI/ML Researcher · Instructor

New York, NY
aalfatemi@fordham.edu
alialfatemi.github.io
Google Scholar

Computer Science Ph.D. Candidate at Fordham University developing robust and efficient machine-learning systems for network security, computer vision, and multimodal intelligence. Research includes lightweight and noise-tolerant DDoS detection, data-efficient learning, LLM-enhanced network analysis, fine-grained visual recognition, and vision–language generation.

EDUCATION

Expected Spring 2027

Ph.D. Candidate in Computer Science · Fordham University

New York, NY · Research focus: AI for network security, robust and data-efficient learning, computer vision, and multimodal AI.

RESEARCH AREAS

- AI for network and cybersecurity
- Robust and data-efficient machine learning
- DDoS detection for edge and IoT systems
- Computer vision and fine-grained recognition
- Large language models for network analysis
- Vision–language and multimodal learning

TEACHING EXPERIENCE

Spring 2026

Instructor · CISC 1100 E01, Structures of Computer Science

Fordham University · Discrete structures, logic, relations, functions, combinatorics, graph theory, and computer-based laboratory work.

Fall 2025

Instructor · CISC 1100 R03, Structures of Computer Science

Fordham University · Set theory, logic, Boolean algebra, recursion, graphs, and rigorous problem solving.

OPEN-SOURCE RESEARCH SOFTWARE

warbler_yolo · YOLO11 segmentation and fine-grained classification pipeline

End-to-end workflow that segments birds from unannotated field images, standardizes foreground crops and data splits, and trains and evaluates a YOLO11 classifier. Supports the peer-reviewed foreground-centric recognition study.

Python · PyTorch · Ultralytics YOLO11 · OpenCV · scikit-learn

INTELLECTUAL PROPERTY

Granted

Inventor · CN113537358B

Cancer subtype identification via multi-omics data integration. Patent record.

PROFESSIONAL SERVICE

Peer-review service: Information Fusion; IEEE Transactions on Network and Service Management; Scientific Reports; Artificial Intelligence Review; Signal, Image and Video Processing; The Journal of Supercomputing; Cluster Computing; IEEE WCCI 2024; IJCNN 2025.

PUBLISHED PAPERS · 22

1. **Ali Alfatemi**, Mohamed Rahouti, Abdellah Chehri, and Zakirul Alam Bhuiyan. “ShallowNet: A Lightweight Neural Network Approach for Efficient Flow-Level DDoS Detection.” IEEE Transactions on Network and Service Management, vol. 23, pp. 5940–5949, 2026. DOI
2. **Ali Alfatemi**, Mohamed Rahouti, Mohammed Aledhari, Nasir Ghani, Abdellah Chehri, and Gwanggil Jeon. “Vision-Language Integration for Image Captioning Using Vision Transformers and GPT-J.” AIPR 2025, LNCS 16446, pp. 101–114, 2026. DOI
3. Mohammad Iqbal Saryuddin Assaqty, Ying Gao, Xiping Hu, **Ali Alfatemi**, Handy Fernandy, Ircham Ali, and Peihao Zhang. “XManuChain: A Permissioned Blockchain Framework for Securing IoP Transactions in Production Process Collaboration.” Journal of Network and Systems Management, vol. 34, issue 4, article 119, 2026. DOI

4. Aiman Solyman, Ahmed Elazab, Mohamed Rahouti, **Ali Alfatemi**, Zeinab Mahmoud, and Marco Zappatore. “MS-GBANet: Multiscale graph convolution with boundary attention for medical image segmentation.” Biomedical Signal Processing and Control, vol. 121, article 110397, 2026. DOI
5. Zeinab Mahmoud, Marco Zappatore, Natalia Kryvinska, Mohammed Abdalsalam, **Ali Alfatemi**, and Aiman Solyman. “MTAGEC: Multi-Task Arabic Grammatical Error Correction as a sequence generation with synthetic data.” Journal of King Saud University Computer and Information Sciences, vol. 38, issue 2, article 35, 2026. DOI
6. Mohammad Iqbal Saryuddin Assaqty, Ying Gao, **Ali Alfatemi**, Mohamed Rahouti, and Abdellah Chehri. “A Resource-Efficient Blockchain With Delegated Fault-Tolerance for Manufacturing Nodes.” IEEE Transactions on Industrial Informatics, vol. 22, issue 5, pp. 4340–4349, 2026. DOI
7. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, and Aiman Solyman. “A Two-Stage LLM-Enhanced DDoS Detection Framework for Next-Generation IoT and Edge Networks.” IEEE GLOBECOM, CISS Symposium, pp. 7–12, 2025. DOI
8. **Ali Alfatemi**, Mohamed Rahouti, Majjed Al-Qatf, and Senthil Kumar Jagatheesaperumal. “Foreground-Centric learning improves robustness in fine-grained visual recognition.” Signal, Image and Video Processing, vol. 19, issue 14, article 1178, 2025. DOI
9. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, Mohammed Aledhari, and Gwanggil Jeon. “Enhancing DDoS Detection for Edge Networks With Noise-Tolerant Shallow Neural Networks in Computational Social Systems.” IEEE Transactions on Computational Social Systems, pp. 1–14, 2025. DOI
10. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Aiman Solyman, and Mohammed Aledhari. “ProtoMAML: A Hybrid Meta-Learning Approach Integrating Prototypical Networks for Data-Efficient DDoS Attack Detection.” IWCMC, pp. 1144–1149, 2025. DOI
11. Henok Wondimu, **Ali Alfatemi**, Mohamed Rahouti, Abdellah Chehri, Paweł Weichbroth, and Nasir Ghani. “Enabling Real-Time, Explainable DDoS Mitigation via On-Premise Large Language Models and Flow Analysis.” Procedia Computer Science, vol. 270, pp. 1390–1399, 2025. DOI
12. **Ali Alfatemi**, Mohamed Rahouti, D. Frank Hsu, Christina Schweikert, Nasir Ghani, Aiman Solyman, and Mohammad I. Saryuddin Assaqty. “Identifying Distributed Denial of Service Attacks Through Multi-Model Deep Learning Fusion and Combinatorial Analysis.” Journal of Network and Systems Management, vol. 33, issue 1, article 8, 2025. DOI
13. Zeinab Mahmoud, Natalia Kryvinska, Mohammed Abdalsalm, Aiman Solyman, **Ali Alfatemi**, and Ahmad Musyafa. “Toward Utilizing Bidirectional Multi-head Attention Technique for Automatic Correction of Grammatical Errors.” IAENG International Journal of Computer Science, vol. 51, issue 11, pp. 1886–1897, 2024.
14. **Ali Alfatemi**, Diogo Oliveira, Mohamed Rahouti, Abdelatif Hafid, and Nasir Ghani. “Precision DDoS Detection Through Gaussian Noise-Augmented Neural Networks.” NoF, pp. 178–185, 2024. DOI
15. Senthil Kumar Jagatheesaperumal, Mohamed Rahouti, **Ali Alfatemi**, Nasir Ghani, Vu Khanh Quy, and Abdellah Chehri. “Enabling Trustworthy Federated Learning in Industrial IoT: Bridging the Gap Between Interpretability and Robustness.” IEEE Internet of Things Magazine, vol. 7, issue 5, pp. 38–44, 2024. DOI
16. **Ali Alfatemi**, Mohamed Rahouti, D. Frank Hsu, and Christina Schweikert. “Advancing NCAA March Madness Forecasts Through Deep Learning and Combinatorial Fusion Analysis.” IntelliSys, LNNS 1067, pp. 539–560, 2024. DOI
17. Fernando Martinez, Mariyam Mapkar, **Ali Alfatemi**, Mohamed Rahouti, Yufeng Xin, Kaiqi Xiong, and Nasir Ghani. “Redefining DDoS Attack Detection Using a Dual-Space Prototypical Network-Based Approach.” ICCCN, pp. 1–9, 2024. DOI
18. **Ali Alfatemi**, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, and Abdellah Chehri. “Multi-Label Classification with Deep Learning and Manual Data Collection for Identifying Similar Bird Species.” Procedia Computer Science, vol. 246, pp. 558–565, 2024. DOI
19. **Ali Alfatemi**, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, Ruhul Amin, and Abdellah Chehri. “Refining Bird Species Identification Through GAN-Enhanced Data Augmentation and Deep Learning Models.” Procedia Computer Science, vol. 246, pp. 548–557, 2024. DOI
20. Zeinab Mahmoud, Chunlin Li, Marco Zappatore, Aiman Solyman, **Ali Alfatemi**, Ashraf Osman Ibrahim, and Abdelzahir Abdelmaboud. “Semi-Supervised Learning and Bidirectional Decoding for Effective Grammar Correction in Low-Resource Scenarios.” PeerJ Computer Science, vol. 9, e1639, 2023. DOI
21. Aiman Solyman, Marco Zappatore, Wang Zhenyu, Zeinab Mahmoud, **Ali Alfatemi**, Ashraf Osman Ibrahim, and Lubna Abdelkareim Gabralla. “Optimizing the Impact of Data Augmentation for Low-Resource Grammatical Error Correction.” Journal of King Saud University – Computer and Information Sciences, vol. 35, issue 6, article 101572, 2023. DOI
22. **Ali Alfatemi**, Hong Peng, Wentao Rong, Bin Zhang, and Hongmin Cai. “Patient subgrouping with distinct survival rates via integration of multiomics data on a Grassmann manifold.” BMC Medical Informatics and Decision Making, vol. 22, article 190, 2022. DOI

PREPRINTS · 5

1. **PREPRINT** Nasim Paykari, **Ali Alfatemi**, Damian M. Lyons, and Mohamed Rahouti. “Integrating Robotic Navigation with Blockchain: A Novel PoS-Based Approach for Heterogeneous Robotic Teams.” arXiv:2505.15954, 2025. DOI
2. **PREPRINT** Mohammed Aledhari, Mohamed Rahouti, and **Ali Alfatemi**. “Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data.” arXiv:2411.05880, 2024. DOI
3. **PREPRINT** Nasim Paykari, Razieh Jokar, **Ali Alfatemi**, Damian Lyons, and Mohamed Rahouti. “Optimizing Control Strategies for Wheeled Mobile Robots Using Fuzzy Type I and II Controllers and Parallel Distributed Compensation.” arXiv:2409.17161, 2024. DOI
4. **PREPRINT** **Ali Alfatemi**, Mohamed Rahouti, Ruhul Amin, Sarah ALJamal, Kaiqi Xiong, and Yufeng Xin. “Advancing DDoS Attack Detection: A Synergistic Approach Using Deep Residual Neural Networks and Synthetic Oversampling.” arXiv:2401.03116, 2024. DOI
5. **PREPRINT** Mohammad Iqbal Saryuddin Assaqty, Ying Gao, **Ali Alfatemi**, Ashraf Osman Ibrahim, Anas W. Abulfaraj, Faisal Binzagr, Fezan Nabawi, and Mohammad Reza Fahlevi. “PBEMS: A Permissioned Blockchain-based Equipment Maintenance System in Smart Manufacturing.” Preprints.org, version 2, 2023. DOI